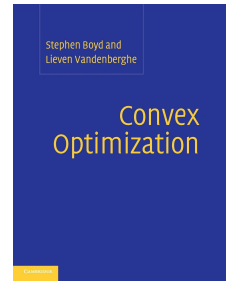


Wearable Robotics

L17 - Using Convex Optimization in the Context of Wearable Robotics

Prof. Bolívar

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Recall

A convex optimization problem follows the form

Minimize $f_0(x)$
subject to $f_i(x) \leq 0, i=1, \dots, m$

where $f_i(x), i=0, \dots, m$ are convex functions

Notice that we used only inequality constraints. If we want to include equality constraints, we can use affine functions

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, i=1, \dots, m \\ & Ax+b=0, A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m \end{array}$$

why?

Definition: A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is **concave** if $-f(x)$ is convex.

Explanation: Any equality constraint $g(x)=0$ can be expressed with two inequalities

$$g(x) \leq 0 \\ 0 \leq g(x) \leq 0 \iff -g(x) \leq 0$$

Thus, for all functions in the inequalities to be convex we must have

$g(x)$ and $-g(x)$ to be convex.

A function that satisfies this condition is

$$g(x) = Ax+b; \text{ Affine function.}$$

Three main strategies to identify if a function is convex

- 1) Check if the function satisfies the definition, i.e., a function is convex if, $\text{dom}(f)$ is a convex set and for any $x, y \in \text{dom}(f)$

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y), \quad 0 \leq \theta \leq 1$$

Its gradient, ∇f , exists at each point in $\text{dom}(f)$

- 2) If the function is differentiable, evaluate the first- or second-order conditions. A function is convex if and only if $\text{dom}(f)$ is convex and

$$f(y) \geq f(x) + \nabla f(x)^T (y-x)$$

first-order condition

or

$$\nabla^2 f(x) \succeq 0$$

Second-order condition

Note: $\succeq \neq \geq$

Curly sign to emphasize that it may refer to a matrix

Second derivative (Hessian)
Is positive semidefinite

- 3) Develop a library of known convex functions. Evaluate if your function of interest results from operations over convex functions that preserve convexity

- Nonnegative weighted sums
- Composition with an affine mapping
- Pointwise maximum and supremum
- Composition
- Minimization
- Perspective of a function

Examples of convex optimization in the context of Wearable Robotics

- Fourier Transform (truncated)
- Fourier + convex function fit
- Model Predictive Control (e.g. D.C. motor) — cvxgen
- signal smoothing (change in muscle cross section)
- Geometric Programming?
- Design of PEA and SEA
- Kalman Filter

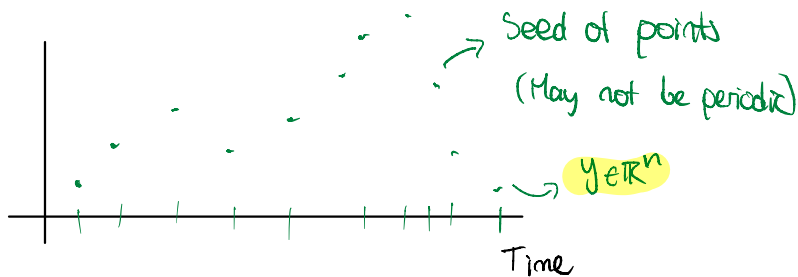
Code available ↗

Examples in the context of Wearable Robotics.

Let's assume you have n -points that describe the trajectory of a joint within a stride (walking, stair, etc)

Q: How can you create a fit to those points that guarantees periodicity?

A: A Fourier transform.



you would like to fit the grid of points using

$$f(t) = \sum_{i=1}^n [a_i \sin(i\omega t) + b_i \cos(i\omega t)]$$

↑
order of the fit.

1) What is the optimization objective?

$\min_{a, b} \|y(t) - f(t)\|_2 \rightarrow$ This is a least-squares problem if we look closely

2) What are the optimization variables?

The coefficients a and b , a set in $\mathbb{R}^{2N}$

3) What are the constraints?

Details

1a) Select fundamental frequency, ω , from application. For example, if you want your signal to be periodic every 2 seconds, $\omega = \frac{2\pi}{2} \text{ (rad/sec)}$

1b) Generate regressor matrix from $f(t)$

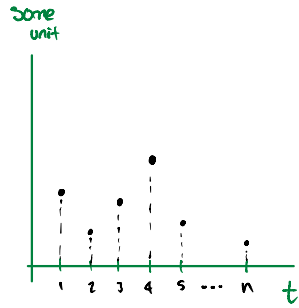
$$M = \begin{bmatrix} \sin(\omega t_1) & \cos(\omega t_1) & \dots & \sin(i\omega t_1) & \cos(i\omega t_1) \\ \vdots & \vdots & & \vdots & \vdots \\ \sin(\omega t_n) & \cos(\omega t_n) & \dots & \sin(i\omega t_n) & \cos(i\omega t_n) \end{bmatrix}; \quad x = \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ a_i \\ b_i \end{bmatrix} \Rightarrow \min_x \|y - Mx\|_2$$

Important affine mappings

First-, second-, and higher-order derivatives can be approximated using affine functions

for example, $x \in \mathbb{R}^n$ then $\dot{x} \approx Dx$

Let's assume the independent variable is time; however, the method applies to any independent variable. Let's say $x \in \mathbb{R}^n$ represents a sampled version of a time trajectory



$\Delta t = t_2 - t_1$ (assume it is constant)

$$\frac{dx}{dt} = \begin{bmatrix} \frac{dx_1}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{bmatrix} \approx \frac{1}{\Delta t} \begin{bmatrix} x_2 - x_1 \\ x_1 - x_2 \\ \vdots \\ x_n - x_{n-1} \end{bmatrix}$$

$$= \frac{1}{\Delta t} \begin{bmatrix} -1 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & -1 & 1 \end{bmatrix} x$$

can be non in some cases, e.g. periodicity

Note: The approximation

$$\frac{dx_i}{dt} \approx \frac{x_{i+1} - x_i}{\Delta t}, \text{ has first-order accuracy} \rightarrow \text{A.K.A forward difference}$$

The approximation

$$\frac{dx_i}{dt} \approx \frac{x_{i+1} - x_{i-1}}{2\Delta t}, \text{ has second-order accuracy. if } \Delta t \text{ is constant.} \rightarrow \text{A.K.A. central difference}$$

Prefer the central difference if the sample time is constant. In our previous example,

$$\dot{x} \approx Dx \Rightarrow D = \frac{1}{2\Delta t} \begin{bmatrix} -1 & 0 & 1 & 0 & \dots & 0 \\ 0 & -1 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & -1 & 0 & 1 \end{bmatrix}$$

(n-2) x n

Can be square in some cases, e.g. periodic motion

For details, see

[canvas/Resources/Handout_Numerical*](#)

Integrals

Integrals can be also approximated using affine functions.

For example:

$$\int f(t) dt \approx \sum_{i=1}^n f(t_i) \Delta t$$

→ Riemann Sum

If $x \in \mathbb{R}^n$ represents a signal sampled at a rate Δt then.

$$x_i = f(\Delta t i)$$

↙ signal

$$\int_0^{b=n\Delta t} f(t) dt \approx \sum_{i=1}^n x_i \Delta t,$$

$$= \Delta t [1 \ 1 \ \dots \ 1] x$$

The cumulative sum is

$$x \approx D \dot{x}$$

$$= \Delta t \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{bmatrix} \dot{x}$$

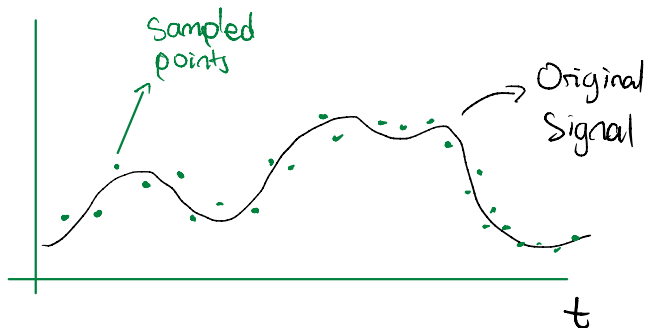
Similar as the central difference, the trapezoidal integration method is preferred when the sample time is constant, (see handout for details)

$$x \approx D \dot{x}$$

$$= \Delta t \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & \dots & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{2} & 1 & 1 & \dots & 1 & \frac{1}{2} \end{bmatrix} \dot{x}$$

Applications

Let's say we sample a signal that is affected by random noise.



How can you estimate the value of the original signal from the noisy measurements?

CVX!

Formulation

$y \in \mathbb{R}^n$: Vector with measurements

$x \in \mathbb{R}^n$: Optimization variable (smooth version of y)

$$\min_x \|y - x\|_2 + \alpha \|Dx\|_2$$

Parameter to change relative importance of tracking v.s. smoothness

An alternative formulation

$$\begin{aligned} \min_x \quad & \|y - x\|_2 \\ \text{s.t.} \quad & \max \|Dx\|_2 \leq \alpha \end{aligned}$$

the same as $\|Dx\|_{\infty}$

Note that constraint functions can be seen as objective functions.

Check example code on the MATLAB drive

Model predictive control

How should we design the control input at sample k , $u_k \in \mathbb{R}^m$, for a discrete-time linear dynamic system

$$\left. \begin{aligned} x_k &= Ax_{k-1} + Bu_k \\ y_k &= Cx_k + Du_k \end{aligned} \right\} \begin{array}{l} \text{can be the} \\ \text{linear model of} \\ \text{our DC-motor} \end{array} ?$$

CVx!

$x \in \mathbb{R}^n$: state of the system

$y \in \mathbb{R}^p$: Outputs of the system

$u \in \mathbb{R}^m$: Inputs of the system

Define the optimization objective

from a known initial condition of the state (x_0) find the optimal control input u_k that guarantees "state" tracking in the future (x_{des})

$$\min_{x, u} \sum_{i=1}^l \left[\|x_{des,i} - x_i\|_2^2 + \alpha \|u_i\|_2^2 \right]$$

Horizon

s.t.

$$y_{lo} \leq u_i \leq y_{hi}, i=1, \dots, l \rightarrow \begin{array}{l} \text{upper bound} \\ \text{of control} \\ \text{low} \end{array}$$

$$x_i = Ax_{i-1} + Bu_i, i=1, \dots, l$$

Define the optimization variables

x : future state

u : future control input.

Check sample code on the MATLAB drive